ASTANA IT UNIVERSITY
School of Software Engineering · Group SE-2413

DIPLOMA PROJECT
FIRST CHECKPOINT REPORT

Development of an Intelligent Voice Control System in the Kazakh Language Based on Neural Networks

Russian title: Разработка интеллектуальной системы голосового управления на казахском языке на основе нейронных сетей

Kazakh title: Нейрондық желілер негізінде қазақ тіліндегі дауыстық басқарудың интеллектуалды жүйесін әзірлеу

Student	Status of assigned focus
Myrzatay Alan Abayuly	ML/ASR study and research (proposed)
Temirgali Nurislam Kairatuly	Backend developer (confirmed)
Abdigali Nurassyl Zhandosuly	Frontend and QA (proposed; confirm with supervisor)

Supervisor: Turarova Marzhan Kabdikalievna

Date: 22 October 2026 | Target deadline: 24 October 2026, 18:00 (Astana local time)

DRAFT STATUS. This report contains planned design and literature research. Interview evidence, approved team allocation, real Git/YouTrack activity, and supervisor feedback must be added before final submission.

Executive summary

This report lays out a feasible design and evaluation plan for an intelligent Kazakh-language voice-command system. The proposed minimal demonstrator accepts controlled voice requests, uses neural speech recognition to create a transcript, identifies one of a limited set of safe intents, and displays a corresponding action outcome. The intended outcome is not merely a speech-to-text demo: users should be able to perform reproducible commands in Kazakh while the team can quantify recognition errors, intent errors, end-to-end latency, and failure behavior. At this checkpoint the team is expected to substantiate the problem, architecture, technologies and next steps; a complete application is not required.

Evidence convention: VERIFIED = backed by a citation or dated evidence; PROPOSED = plan awaiting approval; TODO = user/team must add actual evidence. Claimed model quality, fieldwork, Git activity and supervisor sign-off are not fabricated.

1. Project proposal

1.1 Problem and motivation

Kazakh speech technology is actively researched, including dedicated speech corpora and multilingual training, but existence of a general ASR model does not prove that a Kazakh voice-control interface reliably executes a particular command set under realistic noise, accents and code-switching conditions [S1, S2, S4]. The engineering problem is to connect speech recognition with constrained intent detection, safe action execution, an inspectable backend/API, and reproducible testing.

1.2 Target users, aim and scope

PROPOSED target audience: Kazakh-speaking users who prefer hands-free interaction with a web-based demonstrator; final personas must be validated by genuine stakeholder discussions. Aim: design, integrate and evaluate a safe prototype of neural-network-based Kazakh voice control.

- Collect actual stakeholder evidence and derive user-command priorities.
- Compare Kazakh-capable ASR models and document technology trade-offs.
- Define API, safe actions, component boundaries, and input/output behavior.
- Build a small web demonstrator with a limited, testable intent vocabulary.
- Evaluate ASR quality, command accuracy, latency, errors and privacy controls.

PROPOSED included: audio upload/recording (later), transcript, command inference, allow-listed UI actions, confidence/error handling, metrics. Excluded for MVP: arbitrary OS commands, financial actions, access to real smart-home devices, always-on listening, uncontrolled cloud sharing, and unsupported languages.

1.3 Expected deliverables

Deliverables include a maintained Git repository, traceable project tasks, a documented API, neural ASR integration, web UX, tests, benchmark notebook/scripts, evaluation tables and a final diploma project report. Acceptance at this checkpoint focuses on approved scope, reviewable diagrams, requirements and feasibility, not a completed model.

2. Evidence of relevance — NOT YET VERIFIED

The checkpoint requires 3–5 stakeholder discussions or substantive agreement with a real client. None of the materials available here establish that these have occurred. Do not submit invented transcripts, interview dates or approvals. The team must add real notes to docs/relevance-interview-log-TEMPLATE.md and identify the source of each requirement.

Evidence ID	Stakeholder / method	Verified finding	Impact on specification
INT-01	TODO: real conversation	TODO	TODO
INT-02	TODO: real conversation	TODO	TODO
INT-03	TODO: real conversation	TODO	TODO
INT-04/05	Optional as appropriate	TODO	TODO

Suggested interview focus: preferred app to control, common Kazakh commands, noisy environments, acceptable false activations, language switching, privacy and confirmation before an action. Separate stakeholder statements from team assumptions. If there is a client, document the substantive agreement and scope decisions instead.

3. Critical literature review

Ten scientific publications and five primary technical documentation sources have been selected for meaningful use. Most research sources are from 2021–2024; the 2020 wav2vec 2.0 work is retained for foundational methodology. The matrix below summarizes core relevance, but the team should read the full scientific papers before final claim-by-claim synthesis.

3.1 Synthesis

Kazakh-focused corpora (KSC, KSC2) demonstrate the feasibility of training and evaluating Kazakh ASR, including code-switched material [S1, S2]. Turkic multilingual transfer [S4] and broader cross-lingual pretraining [S6, S7] motivate alternative models, yet published gains cannot be transferred to this specific command corpus without direct experiments. Whisper [S5] offers an accessible multilingual neural baseline, whereas wav2vec 2.0 [S10] explains a family of self-supervised approaches. FLEURS [S8] informs controlled testing but does not replace a speaker-disjoint sample of voice commands. KazakhTTS2 [S9] matters only if spoken responses are included: ASR and TTS should not be conflated.

3.2 Structured literature matrix

ID / year	Source (short)	Contribution to project	Important caveat
S1 / 2021	A Crowdsourced Open-Source Kazakh Spe	Kazakh ASR data; KSC (~332 h).	Mostly prepared speech; command-domain/noise generalization needs testing.
S2 / 2022	KSC2: An Industrial-Scale Open-Source	Larger Kazakh ASR corpus (~1.2k h), including code-switching.	Licensing, size, domain balance and test leakage require review.
S3 / 2022	A study of transformer-based end-to-e	Kazakh-specific transformer-based ASR study.	Reported accuracy depends on corpus and settings; not directly comparable.
S4 / 2023	Multilingual Speech Recognition for T	Joint multilingual Turkic ASR; language transfer.	Multilingual gains may differ for a constrained command set.
S5 / 2023	Robust Speech Recognition via Large-S	Whisper multilingual weakly supervised ASR.	Compute/latency and Kazakh command accuracy must be measured.
S6 / 2022	XLS-R: Self-supervised Cross-lingual	Cross-lingual pretrained speech representations.	Fine-tuning/data and compute effort may exceed MVP budget.
S7 / 2024	Scaling Speech Technology to 1,000+ L	MMS multilingual ASR framework.	Language availability does not imply domain-level reliability.
S8 / 2022	FLEURS: Few-shot Learning Evaluation	Multilingual held-out benchmark with speech tasks.	Read-speech data differ from voice commands.
S9 / 2022	KazakhTTS2: Extending the Open-Source	Kazakh speech synthesis and speech data resources.	TTS is not ASR; indirect relevance until spoken feedback is planned.
S10 / 2020	wav2vec 2.0: A Framework for	Self-supervised pretraining in	Pre-2021 foundational

	Self-Sup	limited-label settings.	reference, not recent state-of-art proof.

3.3 Research/design gap

The project-specific gap is not proving that Kazakh ASR exists; it is showing whether a selectable neural recognizer plus constrained intent parsing achieves reliable voice control in a scoped domain, including safe fallback behavior. The final result must report test counts and measured metrics rather than rely on external papers' accuracy.

4. Existing-system comparison

The systems below are compared against a common set of criteria: scope, Kazakh readiness, offline behavior, action integration, extensibility, privacy/local mode, maintenance status, and relevance. "Not confirmed" reflects a verification gap, not proof that a system cannot support Kazakh. Based on documented product descriptions [D2–D5].

4.1 Comparison matrix A (four shared criteria)

System	Functional domain	Kazakh support	Offline mode	Intent/action integration
Home Assistant Assist	Yes / smart home	Conditional; language pipeline must be tested	Yes, configurable	Home automation
OpenVoiceOS	Yes / general assistant	Not confirmed end-to-end	Yes, component-dependent	Skills/plugins
Rhasspy	Yes / bounded commands	Not listed as turnkey Kazakh profile	Yes	Intent events (HTTP/MQTT/WS)
Picovoice Rhino	Speech-to-intent / bounded	Not in public support list	On-device	Intent/slots

4.2 Comparison matrix B (four further criteria)

System	Customisable	Local/privacy	Maintenance	Fit to Kazakh MVP
Home Assistant Assist	Yes	Yes	Active	High for automation; outside chosen web-demo scope
OpenVoiceOS	Yes	Yes	Active project	High flexibility; integration burden
Rhasspy	Yes	Yes	Archived 2025	Good reference; maintenance risk
Picovoice Rhino	SDK-based	SDK-based	Commercial SDK	Low immediate language fit

Differentiation (proposed): focus tightly on a measurable Kazakh command set with explicit transcript→intent→safe action traceability, rather than promise a general assistant. Rhasspy is included as an informative comparable pipeline but its repository was archived in 2025 [D5].

5. Technology comparison and provisional decisions

The following are options to be experimentally or architecturally assessed, not established winners. Final selection should account for the available team skills/hardware, model licensing, empirical Kazakh performance, privacy, and maintainability.

Decision	2–3 alternatives	Provisional choice	Basis and further verification
Neural speech recognition	Whisper multilingual / Vosk Kazakh / XLS-R fine-tuning	Whisper for first benchmark; Vosk offline comparison; XLS-R optional adapted model	Compare held-out Kazakh WER, latency, memory and license; no winner claimed before experiments.
Backend platform	Go (Gin) / Python (FastAPI) / Node.js (Express)	Go (Gin) API gateway plus Python ML service (provisional)	Matches backend ownership and separates model dependencies; adds inter-service complexity.
Storage	PostgreSQL / SQLite / MongoDB	PostgreSQL only if diagnostic metadata persistence is approved; start without storing raw audio	Relational audit metadata if required; no premature collection of voice recordings.
Component interaction	REST/JSON / WebSocket / gRPC	REST/JSON for short clips; WebSocket considered for streaming	Simple client integration now; streaming upgrade measurable later.

Model-quality decision rule: first build the same held-out evaluation set and test at least two runnable candidate recognizers; then compare WER/CER, exact command accuracy, p95 latency, memory, installation effort, and data/license constraints before selecting a production candidate.

6. Requirements specification

Must = essential acceptance; Should = expected if time permits; Could = future. Functional and non-functional requirements are given testable acceptance criteria. The chosen command taxonomy is provisional pending stakeholder review.

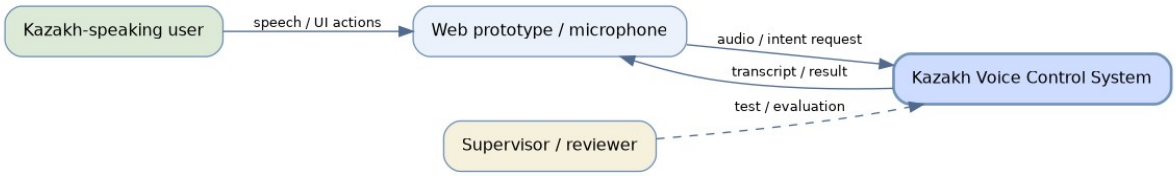
ID	Priority	Requirement	Acceptance criterion
FR-01	Must	Accept Kazakh speech input	Given a supported audio file (WAV 16kHz mono OR documented codec), system accepts upload and returns job/result or validation error.
FR-02	Must	Recognize text with neural ASR	For valid Kazakh audio, response contains transcript and ASR model identifier; no hardcoded sample response in backend.
FR-03	Must	Identify supported command intent	From transcript, map to one of a versioned list of intents or UNKNOWN.
FR-04	Must	Execute safe allow-listed demo actions	Only OPEN_SECTION / SET_THEME / SHOW_HELP / CANCEL can change UI state; unsupported actions do nothing.
FR-05	Must	Present result and errors	Client shows recognized text, intent, action outcome, and understandable error states.
FR-06	Must	Protect data and handle permissions	No microphone access before explicit consent; no raw audio saved by default.
FR-07	Should	Maintain traceable request diagnostics	Store timestamp, model version, latency and outcome without recording sensitive audio (opt-in logs only).

FR-08	Should	Allow text command fallback	Text input invokes intent handler without invoking ASR.
FR-09	Could	Provide spoken Kazakh feedback	Optional TTS only after core command loop is validated.
NFR-01	Must	Evaluation reproducibility	Freeze dataset split, model version, command taxonomy and evaluation script. Metrics reproducible given fixed seed.
NFR-02	Must	Measured latency	Report p50 and p95 end-to-end latency on named machine; target p95 <= 5 seconds for <= 5 s clips (hypothesis, not achieved).
NFR-03	Must	Reliability under invalid input	Malformed/oversize audio yields client error and no server crash; allow-list blocks unsafe execution.
NFR-04	Must	Privacy and security	Credentials in environment variables; audio size limits; consent for any collected recordings.
NFR-05	Should	Kazakh command recognition quality	Evaluate exact-intent accuracy and WER on held-out speakers; targets chosen after baseline pilot.
NFR-06	Should	Maintainability	Model adapter exposes a stable interface; API contract documented with sample requests/responses.

7. Architecture and diagrams

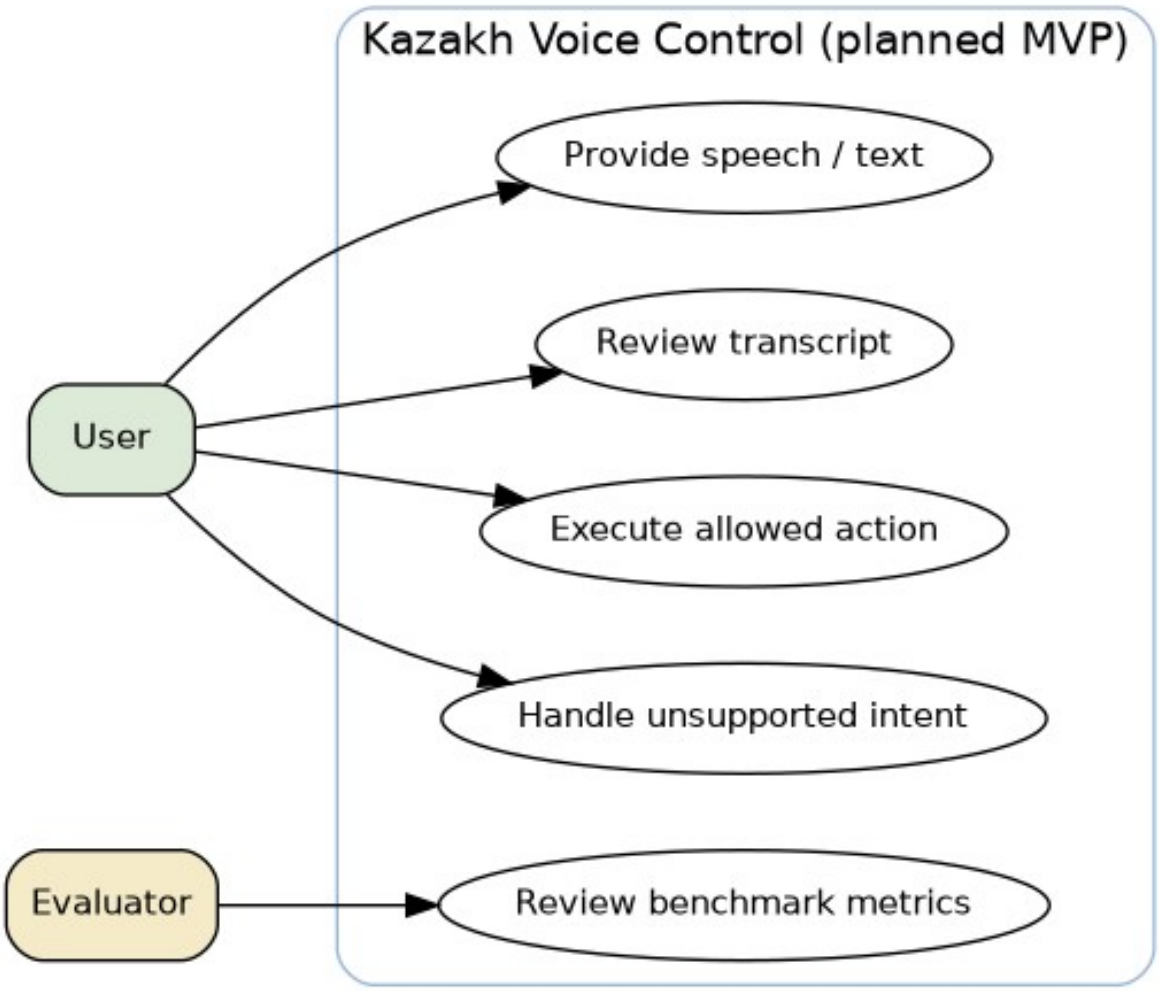
The design separates presentation, backend orchestration, neural model inference, intent/action control and optional minimal metadata storage. This enables model replacement without changing the UI protocol and provides a fail-closed action boundary. Implementation not claimed at this stage.

7.1 System context diagram



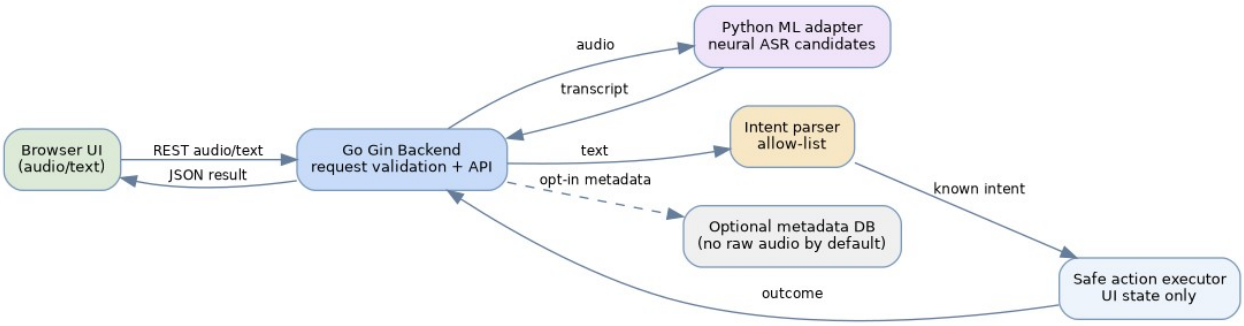
Editable source: diagrams/01-system-context.dot; image export: .svg and .png

7.2 Use case diagram (concept)



Editable source: diagrams/02-use-cases.dot; image export: .svg and .png

7.3 Draft component diagram



Editable source: diagrams/03-component-diagram.dot; image export: .svg and .png

7.4 Proposed interface contract

POST /api/v1/commands/audio accepts a validated short audio file and returns structured JSON containing transcript, intent, action/result, and model information. POST /api/v1/commands/text isolates intent logic during

testing. /health reports service readiness. Invalid formats, oversized audio and model unavailability must have explicit error responses. The detailed examples are in docs/API_CONTRACT_draft.md.

8. UX / prototype

An editable HTML wireframe is included at prototype/index.html. It demonstrates a planned UI flow: enter a sample Kazakh command, show interpreted intent and a safe UI-only effect. The HTML interaction is intentionally mocked. It does NOT record audio or call a neural model yet; it is a UX deliverable, not implementation evidence.

8.1 Concept screen: text-driven static mock

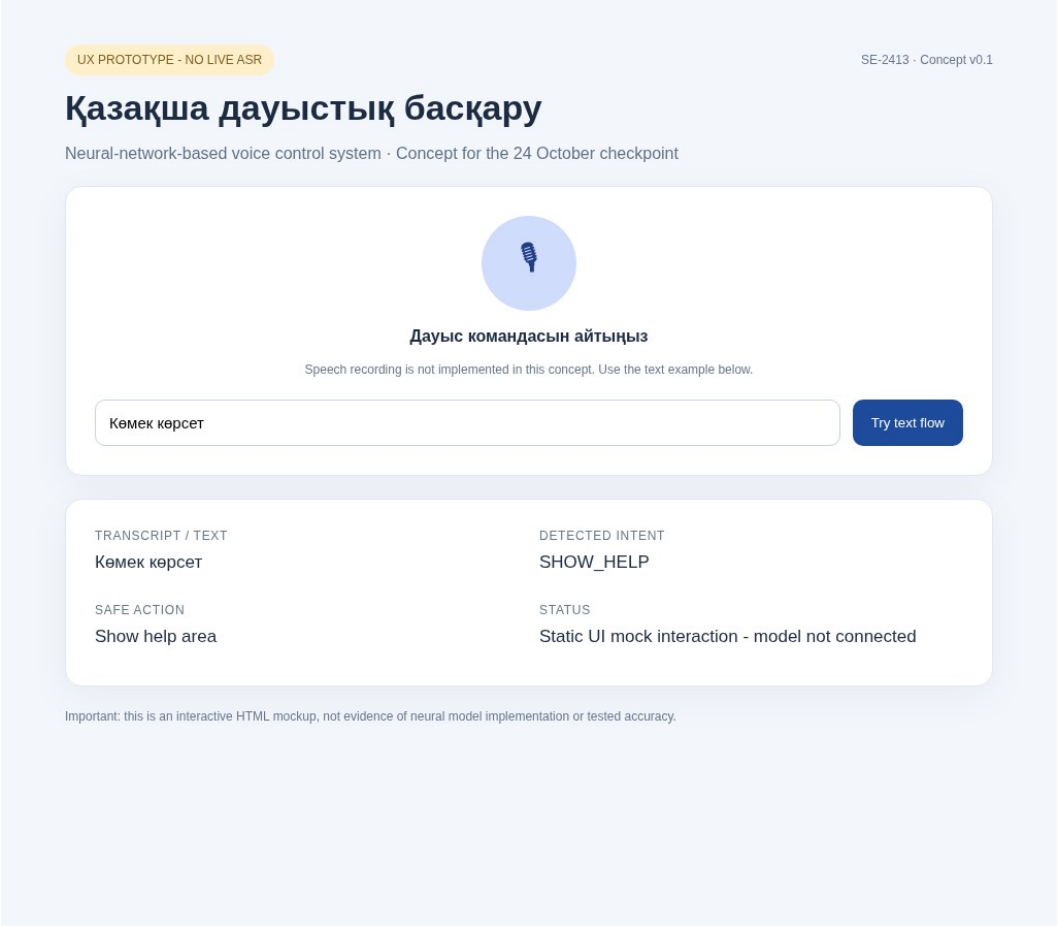


Figure: browser HTML UX mock screenshot. NOT a functioning speech-recognition service.

8.2 Key user flow

- User opens web page and sees permission/privacy note.
- User supplies audio after consent or uses fallback text input.
- System validates input and invokes recognizer (future implementation).
- User sees transcript, recognized intent, feedback and safe action.
- Unknown intent or model error produces a clear message; no unrestricted command executes.

8.3 Prototype acceptance for checkpoint

A reviewer can open the HTML file, input a Kazakh text command, and inspect consistent UI states (supported intent, UNKNOWN, safe theme toggle). A live microphone recording and ASR demonstration are explicitly future development tasks.

9. Verification and evaluation plan

9.1 Protocol

Use consented/duly licensed Kazakh speech clips with documented speaker-disjoint train/development/test separation. Start with public datasets (license verified) and collect a compact domain-specific command evaluation set with consent if approved. Include quiet, background-noise and mixed-language conditions; do not reuse training utterances in held-out test. Maintain fixed model versions and normalization rules.

9.2 Metrics and baseline

Metric	Definition / method	Baseline and status
Word Error Rate (WER)	(S+D+I) / reference words on fixed test clips	Compare candidate models and optional Vosk baseline; NOT measured
Character Error Rate (CER)	Edit distance / reference characters	Track agglutinative Kazakh variations; NOT measured
Intent accuracy	Correct predicted intents / total evaluated commands	Compare keyword/rule baseline to neural-enhanced approach; NOT measured
False action rate	Unsupported commands triggering an action / unsupported commands	Aim zero for known negative cases; NOT measured
Latency p50/p95	End-to-end wall time under named CPU/GPU conditions	Target exploratory p95 <=5 s for <=5 s clips; NOT validated
Robustness	Rate of clean error handling on malformed/empty/oversize input	Acceptance tests to implement; NOT measured

9.3 Planned test cases

Test	Scenario	Expected outcome / coverage
TC-01	Valid Kazakh voice clip	Successful ASR + supported action result [FR-01..05]
TC-02	Russian/Kazakh code-switched clip	Transcript produced, controlled UNKNOWN fallback if ambiguous [FR-02..04]
TC-03	Unsupported phrase	UNKNOWN; no action executed [FR-03..04]
TC-04	Empty/corrupted audio	4xx validation; no server crash [FR-01,NFR-03]
TC-05	Oversized payload	413 or documented 4xx; no processing [NFR-03]
TC-06	Malformed intent from ML service	Fail closed; no unsafe action [FR-04,NFR-03]
TC-07	Microphone access denied	UI error and text fallback path [FR-06,FR-08]
TC-08	Latency test (>= 30 clips)	p50/p95 measured on declared environment [NFR-02]
TC-09	Speaker-disjoint evaluation	WER/CER and intent accuracy by subgroup [NFR-01,NFR-05]

Analysis plan: report total clips, distinct speakers, per-condition samples, metrics for each model under identical conditions, 95% confidence intervals where statistically defensible, error examples, and observed limitations. No claimed performance without real runs.

10. Project management, dependencies and risks

10.1 Team responsibilities — subject to confirmation

Member	Proposed / confirmed focus	Traceable expected artifacts
Nurislam	CONFIRMED backend	API contract, integration design, backend comparison, interface/error specification, model adapter integration later
Alan	PROPOSED ML and literature	Stakeholder evidence coordination, literature matrix, model research/benchmark design
Nurassyl	PROPOSED frontend and QA	UX wireframe, test matrix, user flow, test setup and frontend integration later

Do not use this assignment as a claim of completed work. Actual individual contributions and task activity must be corroborated by members' real commits, issue links and reviewed PRs.

10.2 Near-term milestones

Date	Checkpoint action	Definition of done
22 Oct	Confirm scope; create repo and YouTrack	All member access granted; initial tasks genuinely assigned
23 Oct	Evidence, literature and design review	Real interviews/client evidence; supervisor feedback recorded; corrections tracked
24 Oct before 18:00	Freeze and submit checkpoint	All 11 sections complete; two PDFs; public or approved private links; one Forms response

10.3 Risk register

Risk	Likelihood / impact	Mitigation
R1 Kazakh ASR quality inadequate for natural accents/noise	High / High	Pilot KSC2/FLEURS + original consented test; measure WER and intent accuracy
R2 Not enough genuine stakeholder evidence before checkpoint	High / High	3–5 real interviews or documented client agreement; log dates and approvals
R3 End-to-end latency too high	Medium / High	Benchmark two model sizes; record p50/p95 CPU/GPU
R4 Scope creep into OS-level device control	Medium / High	Explicit limited action allow-list; out-of-scope list and acceptance tests
R5 Accidental exposure of audio/credentials	Medium / High	Opt-in data capture, retention policy, .gitignore, code review
R6 Missing YouTrack/commit evidence at review	High / High	Create real issues, connect PRs, assign owners; stage progress with authentic dates
R7 Supervisor has not reviewed materials	High / High	Schedule review and record exact comments and fixes

10.4 Traceability obligations

Real GitHub or GitLab (GitHub accepted): recognizable individual accounts, README, source/editable diagrams, progressive documentation commits, task-ID branches and peer-reviewed PRs, as required by the checkpoint instructions. Real YouTrack: assigned/priority/deadline tasks, criteria, link to commit/PR, completed only when

acceptance criteria met. The accompanying CSV is an IMPORT TEMPLATE; IDs shown there are not genuine YouTrack IDs.

11. Supervisor feedback — NOT YET VERIFIED

Supervisor: Turarova Marzhan Kabdikalievna. The university requests meaningful review of key outputs and evidence that the team responded to comments. This document contains no verified review meeting or written feedback; the table below must be completed from actual communication.

Date	Deliverable reviewed	Exact feedback	Revision / linked evidence
TODO	TODO	TODO	TODO
TODO	TODO	TODO	TODO

Supervisor confirmation to obtain: approve scope and user population; select MVP action domain and ASR short list; comment on architecture and evaluation; validate responsibilities and repository access.

Submission & integrity checklist

- Finalize stakeholder evidence and separate facts from assumptions.
- Confirm team roles and actual GitHub usernames/individual contributions.
- Have supervisor genuinely inspect artifacts and document comments/revisions.
- Create / populate actual GitHub and YouTrack, ensuring reviewer access.
- Export a separate REAL YouTrack board+task screenshot PDF (not supplied here).
- Replace remaining TODO markers; finalize a single report PDF and submit form once for team before 24 Oct 18:00.

References

[S1] Khassanov et al. (2021). A Crowdsourced Open-Source Kazakh Speech Corpus and Initial Speech Recognition Baseline. https://aclanthology.org/2021.eacl-main.58/

[S2] Mussakhojayeva et al. (2022). KSC2: An Industrial-Scale Open-Source Kazakh Speech Corpus. https://www.isca-archive.org/interspeech_2022/mussakhojayeva22_interspeech.html

[S3] Orken et al. (2022). A study of transformer-based end-to-end speech recognition system for Kazakh language. https://www.nature.com/articles/s41598-022-12260-y

[S4] Mussakhojayeva et al. (2023). Multilingual Speech Recognition for Turkic Languages. https://doi.org/10.3390/info14020074

[S5] Radford et al. (2023). Robust Speech Recognition via Large-Scale Weak Supervision. https://proceedings.mlr.press/v202/radford23a.html

[S6] Babu et al. (2022). XLS-R: Self-supervised Cross-lingual Speech Representation Learning at Scale. https://arxiv.org/abs/2111.09296

[S7] Pratap et al. (2024). Scaling Speech Technology to 1,000+ Languages. https://www.jmlr.org/beta/papers/v25/23-1318.html

[S8] Conneau et al. (2022). FLEURS: Few-shot Learning Evaluation of Universal Representations of Speech. https://arxiv.org/abs/2205.12446

[S9] Mussakhojayeva et al. (2022). KazakhTTS2: Extending the Open-Source Kazakh TTS Corpus With More Data, Speakers, and Topics. https://aclanthology.org/2022.lrec-1.578/

[S10] Baevski et al. (2020). wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations. https://papers.nips.cc/paper_files/paper/2020/hash/92d1e1eb1cd6f9fba3227870bb6d7f07-Abstract.html

[D1] Vosk/Alpha Cephei (n.d.). Vosk offline speech recognition API and models. https://alphacephei.com/vosk/models

[D2] Home Assistant (n.d.). Assist voice assistant overview. https://www.home-assistant.io/voice_control/

[D3] OpenVoiceOS (n.d.). OpenVoiceOS technical manual. https://openvoiceos.github.io/beta-technical-manual/

[D4] Picovoice (n.d.). Rhino Speech-to-Intent documentation. https://picovoice.ai/docs/rhino/

[D5] Rhasspy (2025 archive). Rhasspy GitHub README. https://github.com/rhasspy/rhasspy

Source and status note

This draft was prepared using the AITU checkpoint requirements supplied by the student and the AITU methodology file. Scientific and technical references above provide research context; architecture, acceptance thresholds and proposed roles are proposed project plans, not measured outcomes or supervisor-endorsed facts. The AITU full-thesis guideline regarding length and formatting is separate from this checkpoint submission and should be followed for the final thesis.